



AIR QUALITY INDEX FORECASTING USING GA-KELM

Industrial Oriented Mini Project External Review Presentation

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1 Abstract

- Air quality is a major health concern worldwide.
- A single machine learning model struggles to predict AQI accurately under all conditions.
- We propose GA-KELM — a smarter prediction model using Genetic Algorithm + Kernel Extreme Learning Machine.
- GA-KELM predicts all six major air pollutants: SO₂, NO₂, PM10, PM2.5, O₃, and CO.
- It outperforms existing models — faster training and more accurate predictions.

2 Introduction

- Air pollution is increasing due to industrialization and urbanization, causing around 6.5 million premature deaths annually.
- Air quality is measured using six key pollutants: SO_2 , NO_2 , PM_{10} , $\text{PM}_{2.5}$, O_3 , and CO .
- Machine learning helps predict air quality, but traditional neural networks are slow and may get stuck in local minima.
- GA-KELM model improves performance by combining ELM, kernel functions, and genetic algorithms — making predictions faster, stable, and more accurate.

3 Literature Review

Author(s)	Year	Method	Limitation
Agarwal & Sahu	2023	Statistical models for PM2.5	Not precise for all conditions
Lary et al.	2015	ML + remote sensing data	Limited to PM2.5 only
Zheng et al.	2015	Linear regression + ANN	Cannot capture all nonlinear patterns
Rahput & Sharma	2015	Regression-Based AQI	High computation cost
Jin et al.	2023	Bayesian deep learning	Complex and slow
Xue et al.	2022	GA + Weighted ELM	Not fully optimized

4 Objectives

- To develop a GA-KELM model for accurate and fast AQI prediction.
- To improve prediction using kernel functions for nonlinear data handling.
- To optimize model parameters using a Genetic Algorithm with RMSE as fitness function.
- To predict all major pollutants (SO₂, NO₂, PM₁₀, PM_{2.5}, O₃, CO).
- To build a web application for real-time AQI prediction and visualization.

5 Existing Systems

CMAQ

A physics-based model that simulates air pollutant concentrations. Very accurate but expensive and slow to run.

SVR (Support Vector Regression)

A machine learning method that works well for small datasets but struggles with multi-pollutant, high-dimensional data.

DBN-BP (Deep Belief Network)

Combines deep learning with backpropagation. Powerful but slow to train and sensitive to hyperparameters.

6 Problem Statement

- Air pollution kills millions of people every year — accurate AQI prediction is critical.
- Single ML models fail when AQI patterns change across seasons and weather conditions.
- In ELM, randomly choosing weights and node counts leads to poor predictions.
- KELM performs better — but requires careful tuning of parameters C and γ .
- No existing tool optimizes both discrete and continuous KELM parameters together.
- We need a fast, stable, and accurate AQI prediction system.

7 Functional and Non-Functional Requirements

Functional

- Collect air quality data
- Clean and normalize data
- Accept user input
- Apply GA-KELM model
- Predict AQI and pollutants
- Display results and reports

Non-Functional

- Fast: Trains faster than neural networks
- Accurate: Lower error than baseline models
- Reliable: Consistent across 20 test runs
- Scalable: Works for multiple stations
- Reproducible: Dataset on GitHub

8 Requirement Specification

Software Requirements

- Frontend:React
- Backend:Python ,Fast
- Tools:VS Code
- Models: GA-KELM
- Metrics: RMSE, MSE, R^2

Hardware Requirements

- Computer/Laptop with minimum 8–16 GB RAM
- Multi-core processor (Intel/AMD)
- Stable internet connection

9 Proposed Methodology

- The methodology consists of two main phases:

1. Offline Phase (Model Training)

- Historical dataset is used
- GA-KELM model is trained

2. Online Phase (Real-Time Prediction)

- Live data is fetched from API
- Data is sent to trained model
- AQI is predicted and displayed

10 System Workflow

Step 1: User Interaction

Step 2: Live Data Collection

Step 3: Data Preprocessing

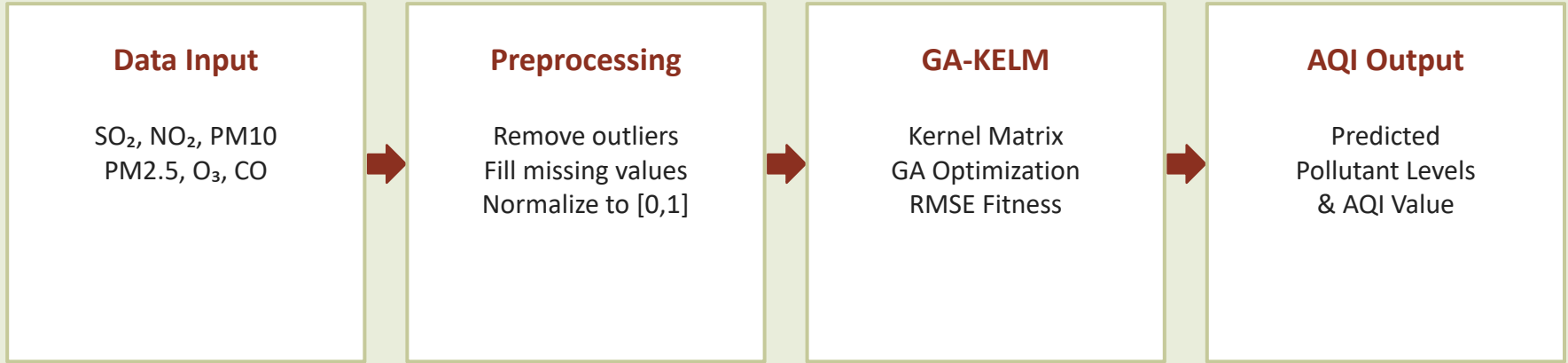
Step 4: Model Prediction

Step 5: Visualization

11 Methodology – GA Optimization of KELM

- Genetic Algorithm (GA) mimics natural evolution to find the best model settings.
- Step 1: Normalize all data
- Step 2: Randomly create a starting population of model settings (weights & thresholds).
- Step 3: Train KELM → test it → calculate RMSE → send to GA as fitness score.
- Step 4: GA selects best settings, applies crossover and mutation to create new solutions.
- Step 5: Repeat until the best result is found (max 40 generations).
- Result: Optimal weights and thresholds are assigned to KELM for final prediction.

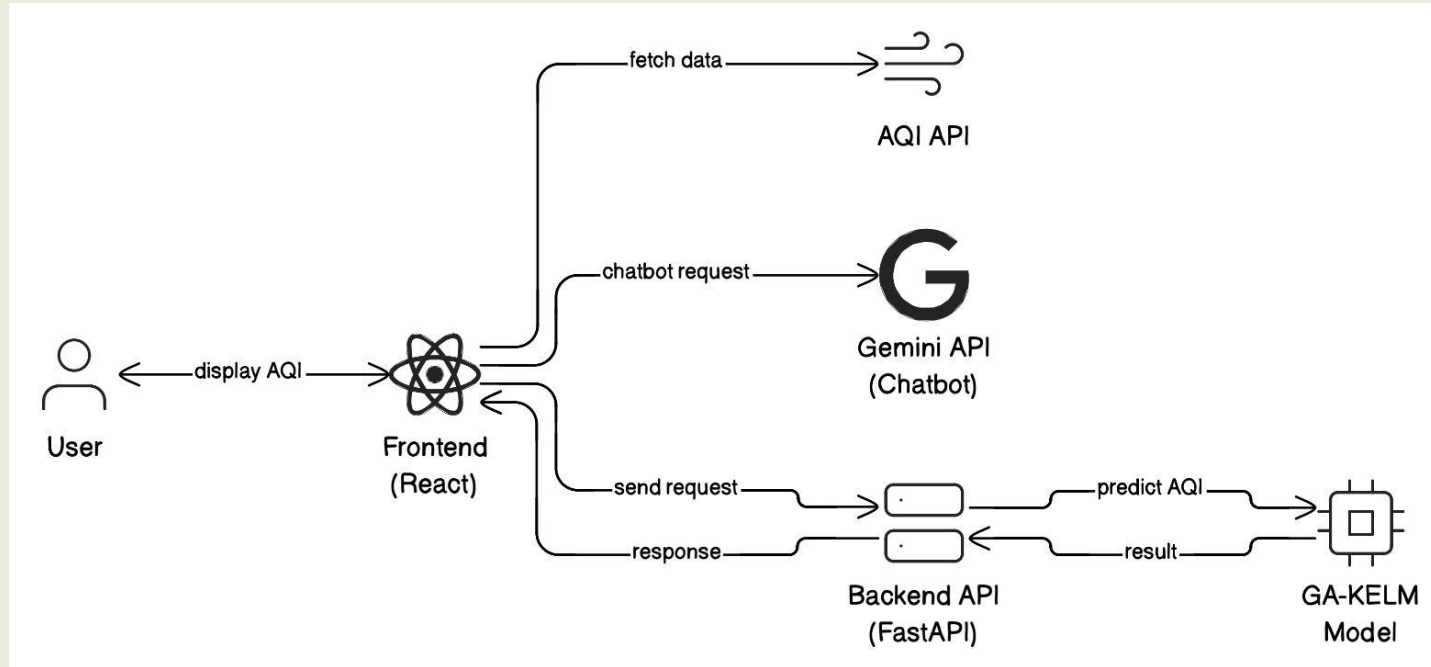
13 Model Architecture



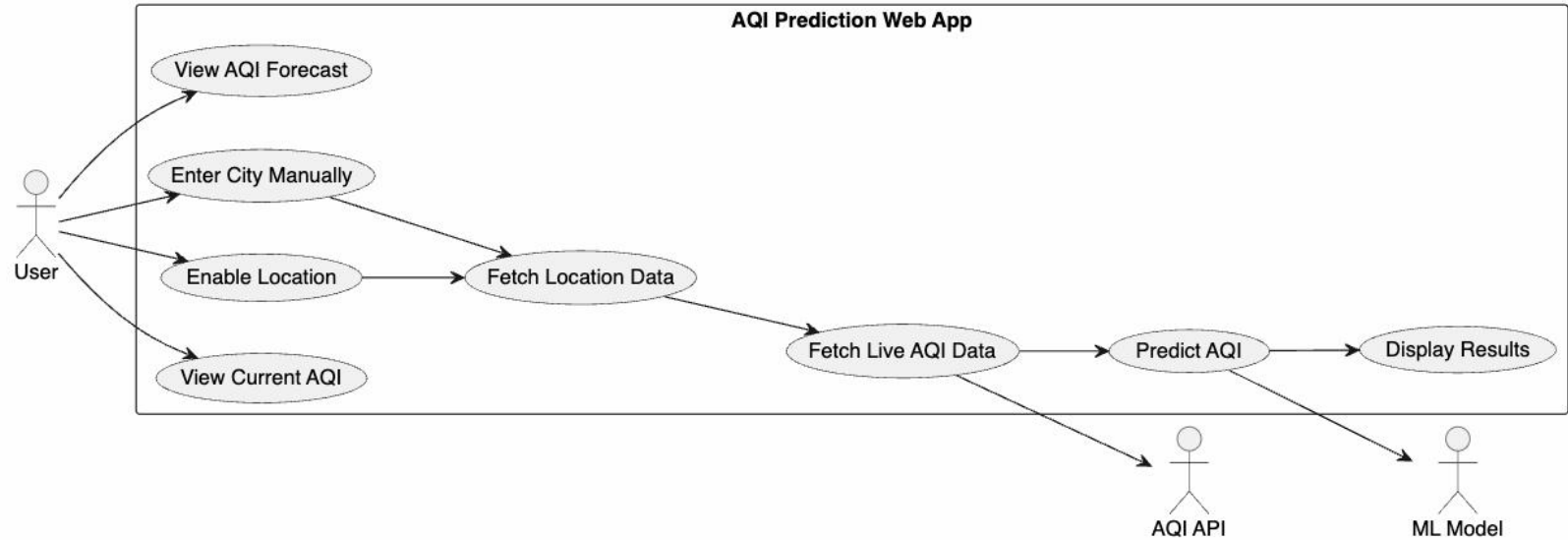
14 Comparision

MODEL	RMSE ↓	MSE ↓	R ² ↑
SVR	18.5	342	0.82
ANN	16.2	290	0.85
KELM	14.8	250	0.88
GA-KELM BEST	10.3	106	0.94

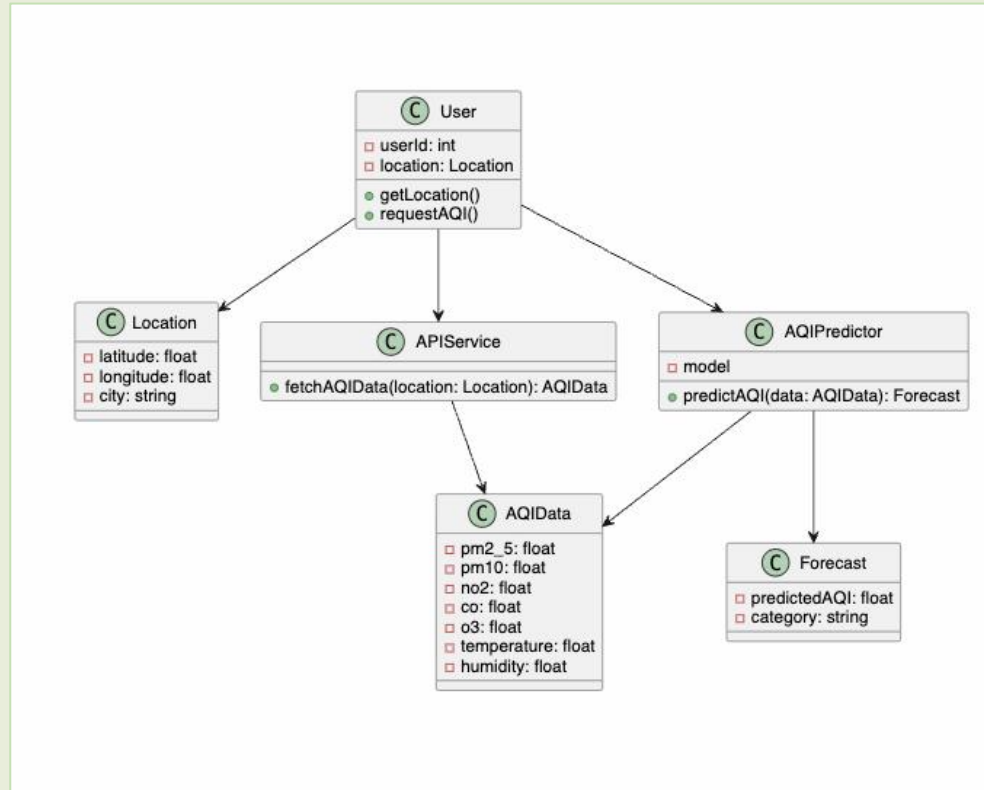
15 SYSTEM ARCHITECTURE DIAGRAM



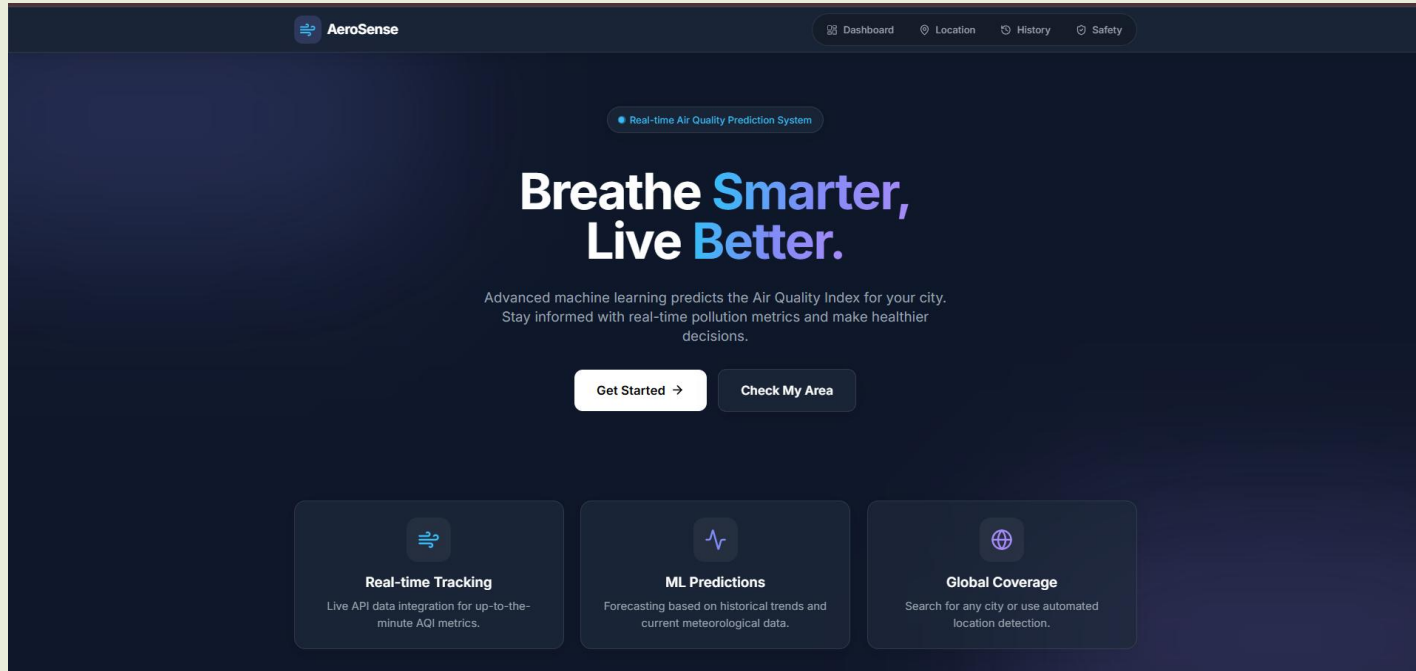
USE CASE DIAGRAM



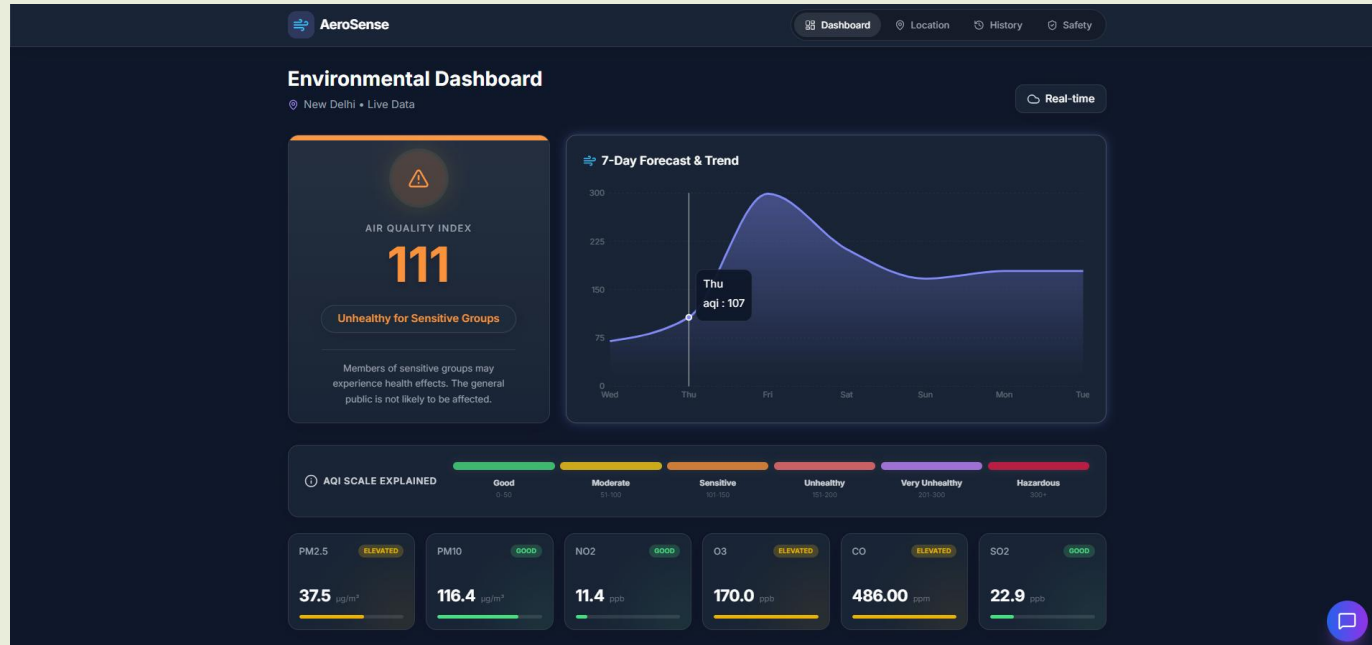
CLASS DIAGRAM



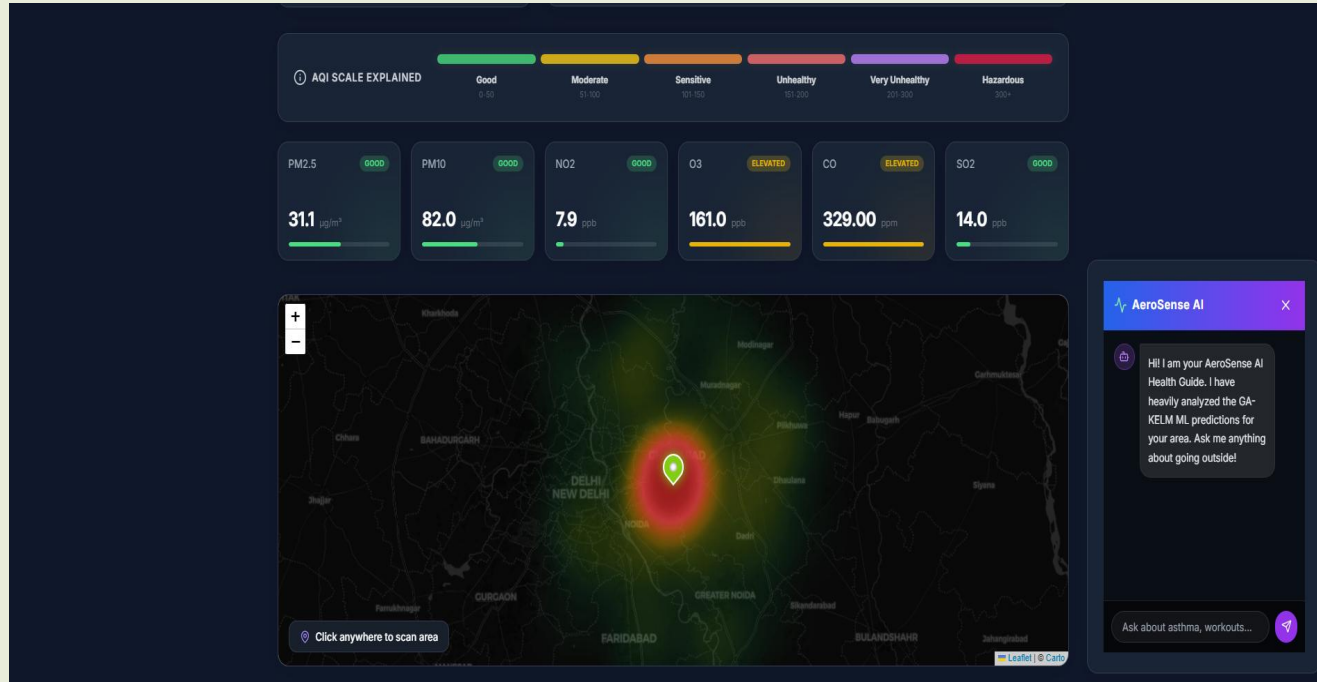
18 Results -Home page



Results -Dashboard



Results -Map & Chatbot



19 Conclusion

- GA-KELM outperforms CMAQ, SVR, and DBN-BP with lower RMSE and higher R^2 .
- Effectively handles complex, nonlinear, and seasonal patterns in air quality data.
- Genetic Algorithm ensures stable and reliable predictions by removing randomness.
- Implemented as a web application for real-time AQI prediction and monitoring.

20 Future scope

- Integrate real-time data from IoT sensors for more accurate and location-specific predictions.
- Develop a mobile application for easy access to AQI information.
- Add alerts and notifications for high pollution levels.
- Integrate with government and environmental monitoring systems for better impact.

21 References

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THANK YOU